Surveillance and GIS for Public-Health Decisions

From reported events to a defensible spatial recommendation

Where should a health office verify, investigate, allocate resources, and communicate?

20-30 minute teaching module | Evidence baseline checked 10 August 2026

What you will be able to do

Distinguish surveillance
from research by purpose

Evaluate data quality
and system performance

Compare counts, rates,
and geographic choices

Recommend action
without causal overreach

Roadmap: four moves from data to decision

1. PURPOSE
What decision?

→

2. QUALITY
How produced?

3. PLACE
What changes by map?

→

4. ACTION
What is proportionate?

The case hook

A fictional regional health office receives four weeks of respiratory-event reports from six service areas.

Leadership asks: **Where is the burden? Where is the reporting problem? What can we show publicly?**

Your job is not to diagnose a community from a map. It is to decide what the evidence can support now—and what should happen next.

First participation prompt

OBSERVATION
What was recorded?

INTERPRETATION
What might it mean?

CAUSAL CLAIM
Why did it happen?

ACTION
What should we do next?

Turn to a partner: give one sentence in each category.

Surveillance is a decision system

Define → Collect → Check

Analyze → Communicate → Act

Feedback closes the loop: information must return in a useful form.

Surveillance or research? Start with purpose

SURVEILLANCE / PROGRAM	RESEARCH
Detection, control, response, or improvement	Develop or contribute to generalizable knowledge

The data source alone does not decide. Purpose, design, intended use, and governance matter.

A count has a production history

Case definition	Who qualifies as a reported event?
Reporting opportunity	Who can enter the system—and who cannot?
Transmission	What delay, loss, or duplication occurs?
Analytic rule	What cleaning, grouping, or threshold changes the total?

A changing count may reflect changing disease, changing detection, or both.

Evaluate what matters for this decision

Simplicity	Flexibility	Data quality
Acceptability	Sensitivity	Predictive value positive
Representativeness	Timeliness	Stability

There is no universal passing score. System objectives set priorities.

Tradeoffs are normal

FASTER
may be less complete

MORE SPECIFIC
may miss events

MORE DETAIL
may reduce acceptability or privacy

The synthetic six-area dataset

Area	Population	Events	Complete	Late
Northgate	52,000	78	94%	7%
Riverbend	18,500	42	82%	24%
Central	81,000	96	97%	4%
Lakeside	9,200	6	76%	18%
Hillview	31,500	60	90%	9%
Southpoint	4,800	< 5	88%	11%

All names and values are synthetic instructional data. They are not official data. < 5 is a classroom suppression rule, not a universal threshold.

Counts and rates answer different questions

CENTRAL: 96 events

Highest workload

RIVERBEND: 22.7 per 10,000

Highest displayed rate

Both can be correct priorities—for different decisions.

Calculate one rate—then audit it

$$\frac{42 \text{ reported events}}{18{,}500 \text{ people}} \times 10{,}000 = 22.7$$

Do numerator, denominator, geography, and period describe the same population?

A signal can also be a data-quality warning

RIVERBEND
22.7 per 10,000
82% complete
24% late

Before escalation:

- Verify the workflow change
- Reconcile late reports
- Check denominator fit
- Examine persistence over time

Highest displayed rate does not mean confirmed highest underlying risk.

Geocoding links records to place

Address → Actual or calculated coordinate → Associated geography

Record match status, approximation, unmatched records, benchmark/vintage, and permitted handling. Address-range coordinates may be interpolated.

Geographic units are analytic choices

County

Census tract

ZIP-related area

Service district

Program-defined area

Administrative convenience is not epidemiologic truth.

A boundary file is not health data

Boundary + vintage + Health numerator + Population denominator

Units and time must be compatible before mapping.

The same values can tell different visual stories

Raw-count choropleth	May mostly display where more people live.
Rate choropleth	Improves comparison, but does not cure denominator error or small-number instability.
Quantile classes	Balance areas across colors, but class limits depend on this dataset.
Meaningful breaks	Support a decision only when the thresholds have a defensible basis.

Show the classification rule and underlying values. Color does not create evidence.

Scale changes both usefulness and risk

LARGER AREAS	FINER AREAS
More stable	More local
More private	Less stable
Less local detail	Greater disclosure risk

Choose the minimum geographic detail needed for the decision.

A spatial pattern is a question, not a cause

Area-level pattern $\neq$ individual cause

A pattern can guide verification, outreach, sampling, or study design.

Avoid ecological fallacy: an area characteristic and area rate do not establish an individual exposure–outcome relationship.

Privacy changes the analysis

SUPPRESSED
Information withheld

ZERO
Observed value of none

Do not infer, back-calculate, or shade a suppressed value as zero.

Fine geography, rare combinations, and repeated releases can disclose more than any one table.

Team decision: seven minutes

ANALYST
Calculate and audit

SKEPTIC
Challenge inference

DECISION LEAD
Choose next action

Produce: one immediate action, one verification step, one additional data element, and one privacy-aware map specification.

Be ready to defend why the action is proportionate to the evidence.

The three-sentence decision brief

The data show... observation with count/rate, place, and period.

The data do not establish... uncertainty, missingness, and causal limit.

We recommend next... proportionate action plus verification.

Key takeaways

01 Start with the public-health decision and purpose.

02 Audit how the data were produced before interpreting the pattern.

03 Align numerator, denominator, geography, period, and boundary vintage.

04 Treat mapping, classification, scale, and suppression as analytic choices.

05 Match the strength of action to the strength of evidence.

Sources and currentness

Official sources used

- CDC, Updated Guidelines for Evaluating Public Health Surveillance Systems.
- CDC, Field Epidemiology Manual: Describing Epidemiologic Data.
- CDC, Environmental Public Health Tracking.
- CDC, CDC WONDER Frequently Asked Questions.
- U.S. Census Bureau, TIGER Data Products Guide.
- U.S. Census Bureau, Census Geocoder Documentation.

Baseline checked 10 August 2026. Verify disease-specific rules, data releases, service status, geographic vintages, denominators, and privacy policies at point of use. The case and data are synthetic.